

Term 1 Week 5

$$1. \int \left(\frac{1-\sin x}{1+\sin x} \times \frac{1-\sin x}{1-\sin x} \right) dx$$

$$\int \frac{1-2\sin x+\sin^2 x}{1-\sin^2 x} dx$$

$$\int \frac{1-2\sin x+\sin^2 x}{\cos^2 x} dx$$

$$\int \frac{1}{\cos^2 x} dx - \int \frac{2\sin x}{\cos^2 x} dx + \int \frac{\sin^2 x}{\cos^2 x}$$

$$\int \sec^2(x) dx - \int \frac{2\sin x}{\cos x} \times \frac{1}{\cos x} dx + \int \tan^2(x) dx$$

$$\int \sec^2(x) dx - \int 2 \tan(x) \sec(x) dx + \int (\sec^2(x) - 1) dx$$

$$= \tan(x) - 2 \sec(x) + \tan(x) - x + c$$

$$= 2 \tan(x) - 2 \sec(x) - x + c$$

2. Using Pythagoras:

$$\sin^2 x + \sin^2 2x = \sin^2 3x$$

We need to rewrite each term so that they are all in terms of $\sin x$

$$\sin^2 2x = (\sin 2x)^2 = (2 \sin x \cos x)^2 = 4 \sin^2 x \cos^2 x$$

$$= 4 \sin^2 x (1 - \sin^2 x)$$

$$= 4 \sin^2 x - 4 \sin^4 x$$

$$\sin 3x = \sin(2x + x) = \sin 2x \cos x + \cos 2x \sin x$$

$$= 2 \sin x \cos^2 x + (1 - 2 \sin^2 x) \sin x$$

$$= 2 \sin x (1 - \sin^2 x) + \sin x - 2 \sin^3 x$$

$$= 2 \sin x - 2 \sin^3 x + \sin x - 2 \sin^3 x$$

$$= 3 \sin x - 4 \sin^3 x$$

$$\text{Therefore, } \sin^3(3x) = (3 \sin x - 4 \sin^3 x)^3 = 16 \sin^6 x - 24 \sin^4 x + 9 \sin^2 x$$

Substituting everything back into the original equation:

$$\sin^2(x) + 4 \sin^2(x) - 4 \sin^4(x) = 16 \sin^6(x) - 24 \sin^4(x) + 9 \sin^2(x)$$

$$16 \sin^6(x) - 20 \sin^4(x) + 4 \sin^2(x) = 0$$

Simplifying and factorising:

$$4 \sin^6(x) - 5 \sin^4(x) + \sin^2(x) = 0$$

$$\sin^2(x)(4\sin^4(x) - 5\sin^2(x) + 1) = 0$$

$$\sin^2(x)(4\sin^2(x) - 1)(\sin^2(x) - 1) = 0$$

Difference of two squares:

$$\sin^2(x)(2\sin(x) + 1)(2\sin(x) - 1)(\sin(x) + 1)(\sin(x) - 1) = 0$$

From here, we just examine each factor and see which gives valid solutions for our triangle.

$$\sin^2(x) = 0$$

$$x = 0 \text{ (Not valid)}$$

$$2\sin(x) + 1 = 0$$

$$\sin(x) = -\frac{1}{2} \text{ (Not valid)}$$

$$2\sin(x) - 1 = 0$$

$$\sin(x) = \frac{1}{2}$$

$$x = \frac{\pi}{6} \text{ (valid)}$$

$$\sin(x) + 1 = 0$$

$$\sin(x) = -1 \text{ (Not valid)}$$

$$\sin(x) - 1 = 0$$

$$\sin(x) = 1$$

$$x = \frac{\pi}{2} \text{ (Not valid as the side with } \sin(2x) \text{ will have length of } \sin(\pi) = 0)$$

Therefore, the only valid solution is $x = \frac{\pi}{6}$.

3. Separate the RHS into two factors:

$$x^x = 5^x \times 5^{25}$$

Divide by 5^x :

$$\frac{x^x}{5^x} = 5^{25}$$

Simplifying the LHS:

$$\left(\frac{x}{5}\right)^x = 5^{25}$$

Take the fifth root:

$$\left(\frac{x}{5}\right)^{\frac{x}{5}} = 5^5$$

Therefore, $\frac{x}{5} = 5$

$$x = 25$$